Robot Run Scoring Worksheet

Full 2:30 test runs. Run it like the real match. Write down every run.

A full test run is a practice match: all six missions in order, with the **2 minute 30 second** timer running. Each run gets its own run sheet. After a few runs, the summary pages show our **average, best, worst and middle score**, and which missions fail most often. That tells us what to fix first, and it is proof for the Robot Design judges.

Our mission order

#	Mission	What the robot does
1	**Mission 9: Research Platform**	
2	**Mission 2: Drone Survey**	
3	**Mission __: Exploding Seeds**	
4	**Mission 15: Biocentric Architecture** (city dock: garden skylight and nesting canopy)	
5	**Mission 6: Leafcutter Frenzy**	
6	**Mission 11: Window to the Past**	

Fill in the right-hand column once, from the rule book, so everyone knows what "done" means for each mission.

Jobs for each run

1. **Driver** runs the robot and swaps attachments. **Timer** starts the 2:30 clock and calls out the time. **Scorer** fills in the run sheet. Swap jobs every run.
2. **Reset the field** exactly like the start of a match, with every model in its starting spot.
3. **Start the timer and go.** When each mission ends, the timer calls out the time left and the scorer writes it in the **Timer** column.
4. **Stop at 0:00** even if missions are left. Missions not tried count as **Failed**, 0 points.
5. **Score each mission** using the rule book, add up the total, and agree on the **one change** to make before the next run.

✓ Done
Did everything the mission asks and got full points.

½ Partial
Got some points but not all of them.

✗ Failed
0 points: missed, broke, ran out of time or not tried.

Rules that keep the numbers honest

- **Never skip a bad run.** If we only write down good runs, our average lies to us.
- **Change one thing between runs** (code, attachment or how we line up), so we know what made the difference.
- Write down every time a hand touches the robot outside home. In a real match that can cost points.

Print one run sheet (page 2) for every run. Pages 3 and 4 collect all the runs together.

Run Sheet Run # _____

Date: _____ **Driver:** _____ **Timer:** _____ **Scorer:** _____

#	Mission	Result	Points	Timer at end	What happened? What should we change?
1	MISSION 9 **Research Platform**	☐ ✓ Done ☐ ½ Partial ☐ X Failed	pts	m : ss left	
2	MISSION 2 **Drone Survey**	☐ ✓ Done ☐ ½ Partial ☐ X Failed	pts	m : ss left	
3	MISSION ___ **Exploding Seeds**	☐ ✓ Done ☐ ½ Partial ☐ X Failed	pts	m : ss left	
4	MISSION 15 **Biocentric Architecture** City dock: garden skylight and nesting canopy	☐ ✓ Done ☐ ½ Partial ☐ X Failed	pts	m : ss left	
5	MISSION 6 **Leafcutter Frenzy**	☐ ✓ Done ☐ ½ Partial ☐ X Failed	pts	m : ss left	
6	MISSION 11 **Window to the Past**	☐ ✓ Done ☐ ½ Partial ☐ X Failed	pts	m : ss left	

TOTAL POINTS	TOTAL TIME USED	✓ DONE	½ PARTIAL	X FAILED	HAND TOUCHES
_____ pts	_____ m : ss	_____ of 6	_____ of 6	_____ of 6	_____ times

Total time used = 2:30 minus the timer when the robot finished its last mission. If the timer hit 0:00, write 2:30.

The ONE change for the next run

_____

_____

Who will do it: _____ **Did it help? (next run)** _____

All Runs: Points and Time

Copy the totals from each run sheet into one row. The first row is an example.

Run	Date	Total points	Time used	✓ Done	½ Partial	✗ Failed	Hand touches	The change we made after this run
eg	*11/12*	*145*	*2:22*	*3*	*2*	*1*	*1*	*Slow down the turn before Mission 6*
1								
2								
3								
4								
5								
6								
7								
8								
9								
10								
11								
12								

Our points numbers

Use only the **Total points** column, and don't count the example row.

Average

Add up all the totals, then divide by the number of runs.

_____ pts

Max (best)

The biggest total.

_____ pts

Min (worst)

The smallest total.

_____ pts

Median (middle)

Put the totals in order. The middle one. Even number of runs? Average the two middle ones.

_____ pts

Work space: totals in order, smallest to biggest

What the numbers tell us: is the average close to the max? If not, which missions keep dragging it down?

Why the median? One really bad run (robot fell over) pulls the average down a lot but barely moves the median. If the average is much lower than the median, we have a few disaster runs to look at.

Mission by Mission

For each run, write the mission's result: ✓, ½ or ✗, and its points underneath if there is room. Then count across. The mission with the most ✗ is the one to fix first.

Mission	1	2	3	4	5	6	7	8	9	10	11	12	✓ count	✗ count
1 · Mission 9 Research Platform														
2 · Mission 2 Drone Survey														
3 · Mission __ Exploding Seeds														
4 · Mission 15 Biocentric Architecture														
5 · Mission 6 Leafcutter Frenzy														
6 · Mission 11 Window to the Past														

What to fix next

#	Mission	Why does it fail or only get partial points?	Fix and who
1			
2			
3			

Time check

Slowest mission (biggest time gap on the run sheets): _____

Average time used: _____ of 2:30

Spare time we could use for another mission (2:30 minus average time used): _____

The time a mission takes is the gap between the timer reading before it and after it. Mission 1 starts at 2:30.